MEASURE WHAT MATTERS.

Inspectrum.C

Documentation

Inspectrum.C Documentation

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1. Sicherheitshinweise (deutsch)

Lesen Sie die folgenden Sicherheitshinweise aufmerksam durch, bevor Sie das Produkt in Betrieb nehmen. Folgen sie den Anweisungen, um Verletzungsrisiken für das Bedienpersonal und Beschädigungen des Produkts zu vermeiden. Die Sicherheitsvorkehrungen müssen während des gesamten Systembetriebs beachtet werden.

Symbole und Konventionen in diesem Dokument

Es besteht allgemeine Gefahr, die Personen- oder Sachschaden zur Folge haben kann, wenn Sie die hier angegebenen Anweisungen nicht beachten.

Es besteht die Gefahr von Elektrolytaustritt aus einer Batteriezelle

Es besteht Gefahr durch feuergefährliche Stoffe

Bestimmungsgemäßer Gebrauch

Das Produkt darf ausschließlich für Entwicklungs-, Forschungs- und Testzwecke sowie im Produktionsumfeld eingesetzt werden. Der unsachgemäße Einsatz oder nicht bestimmungsgemäße Gebrauch ist nicht zulässig.

Haftung

Die Bedienung von SAFION Produkten obliegt ausschließlich speziell geschultem und ausgebildetem Personal. Die SAFION GmbH übernimmt keine Haftung für Sach- oder Personenschäden, die aus dem unsachgemäßen oder nicht bestimmungsgemäßen Gebrauch dieses Produktes oder der falschen Bedienung durch unzureichend qualifiziertes Personal resultieren.

Qualifikationen des Anwenders

Es dürfen ausschließlich ausgebildete Elektrofachkräfte am Produkt und den angeschlossenen Komponenten arbeiten oder unterwiesene Personen unter Aufsicht und Anleitung einer ausgebildeten Elektrofachkraft und unter Einhaltung der im Elektrobereich geltenden Vorschriften und Normen. Eine ausgebildete Elektrofachkraft verfügt sowohl über eine technische Ausbildung, Fachwissen und Erfahrung als auch über Kenntnisse relevanter Vorschriften, um die ihm zugewiesenen Aufgaben beurteilen und damit einhergehende Gefahrenpotenziale erkennen zu können.

2. Safety Instructions (english)

Read the following safety instructions carefully before operating the product. Follow the instructions to avoid the risk of injury to the operating personnel and damage to the product. The safety precautions must be observed during the entire system operation.

Symbols and conventions in this document

There is a general danger that can result in personal injury or property damage if you do not follow the instructions.

There is a risk of electrolyte spillage from a battery.

There is danger due to flammable substances

Intended use

The product may only be used for development, research and test purposes, as well as in a production environment. Improper use or use not in accordance with the intended purpose is not permitted.

Liability

The operation of SAFION products is the exclusive responsibility of specially trained and qualified personnel. SAFION GmbH accepts no liability for damage to property or personal injury resulting from improper use of this product or use for other than the intended purpose or incorrect operation by insufficiently qualified personnel.

Qualifications of the user

Only qualified electricians may work on the product and the connected components or instructed persons under the supervision and guidance of a qualified electrician and in compliance with the regulations and standards applicable in the electrical field. A qualified electrician has technical training, specialist knowledge, and experience as well as knowledge of relevant rules to be able to assess the tasks assigned to him and to recognize associated potential dangers.

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3. Specifications

Parameter Inspectrum.C	Value
AC Current Output (1 Hz –10 kHz)	Max. 10 A (peak-peak)
AC Current Output (1 mHz – 1 Hz)	Max. 5 A (peak-peak)
AC Excitation	parallel multi-sine operation, up to 32 frequency points
DC Current Output (charge and discharge option)	max. +/-1 A
EIS measurement charge/discharge	Yes*
EIS measurement during temperature change	Yes*
Temperature Measurement Input	1 CH RTD PT-100 (4-wire)
DC Voltage Range	0 - 5 V
Frequency Range	10 mHz - 10 kHz**
Frequency Accuracy	100 ppm
Impedance Range	0.1 mΩ to 100 mΩ
Interfaces	USB, CAN (free configurable)
Voltage Input Resolution (DC)	< 0.5 mV
Voltage Input Resolution (AC)	< 2 µV
Accuracy |Z| (typical, after calibration)	< 1 %
Accuracy arg(Z) (typical, after calibration)	< 0.5 °
Input Power	< 60 W
Input Voltage	12 VDC
Dimensions (L x W x H)	240x165x55mm
Ambient Temperature	+10°C to+30°C

Table 1: Specifications

(*) An EIS measurement can be performed in parallel during charging and discharging or temperature change tests, but care must be taken to select the lowest frequency and the excitation periods so that the state of the battery does not change significantly during the EIS measurement.

(**) The AC excitation frequency limits the AC excitation current. In the range of 1 Hz – 10 kHz the maximum current is limited to 10 A(peak-peak) and in frequencies below <1 Hz the maximum current is limited to 5 A(peak-peak).

It is extremly important to never connect a battery or other device-under-test (DUT) to the inputs of the Inspectrum.C, whose DC-Voltage is outsite of the in Table 1 specified voltage range. Any mistreatment may cause severe damage to either the device or the battery and will void the warranty.

4. System Overview

Figure 1: Front and back view of the Inspectrum.C

Number	Description
1	Power supply connector
2	Power on/ off pushbutton
3	Device status LEDs
4	Current Excitation output socket
5	Voltage Measurement input socket
6	Temperature sensor socket
7	Multiplexer Interface
8	USB Interface
9	CAN Interface and CAN status LEDs

Table 2: Overview connections and interfaces

5. Hardware Setup and Connections

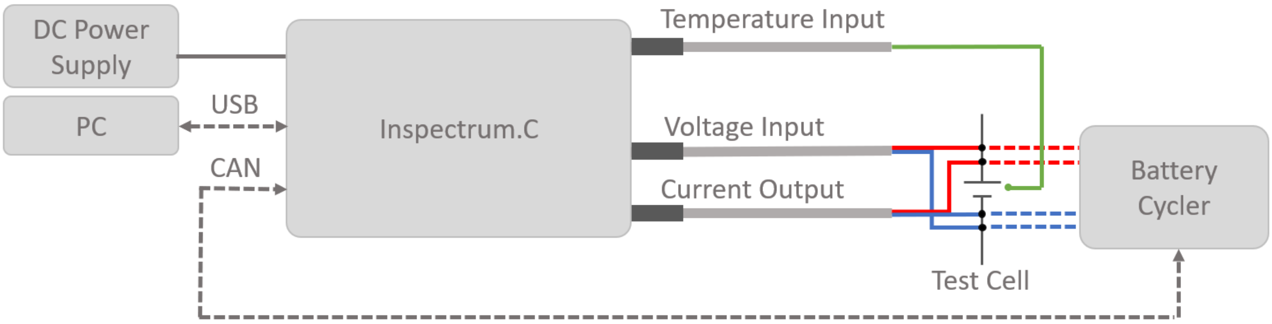


Figure 2: Diagram of the general connection setup with a battery cycler

Figure 2 shows all connections with the device. In order to take the Inspectrum.C into operation, the provided power supply (MeanWell GSM120A12-R7B) must be plugged into the DC power supply connector (1) as shown in Figure 3. If done correctly, the connector is locked into place and can only be removed by pulling back the black sleave of the connector. The general setup and control of the Inspectrum.C is done via the USB interface. Thus, the included USB Cable needs to be plugged into the USB Interface (7) and connected to a host computer or laptop on the other end. After the device configuration is finished, the Inspectrum.C can also be controlled via a CAN interface so that, for example, a battery cycler can be integrated to take over the control. Therefore, the Inspectrum.C needs to be connected to the CAN-BUS via the CAN-Interface (9). It doesn't matter, which one of the two CAN-Bus connectors is chosen, since both ports are connected in parallel internally. Nevertheless, the BUS-Termination needs to be provided externally by the user if necessary.

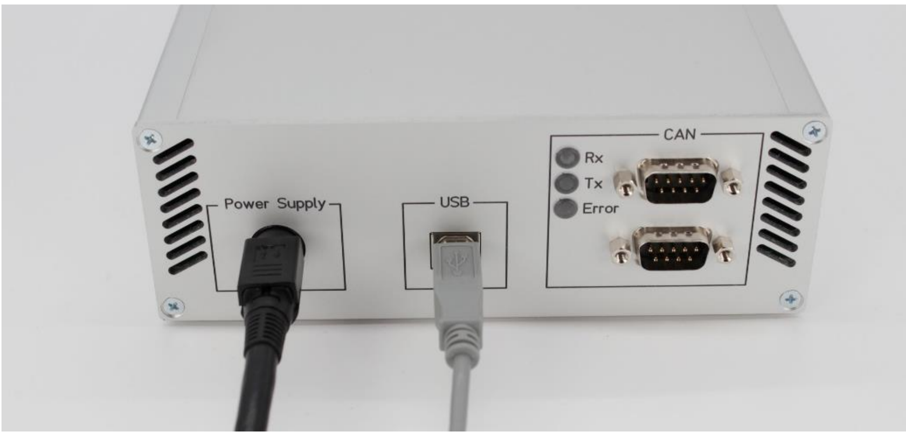


Figure 3: Connection of the power supply and USB-Interface

The battery cell needs to be connected to the current excitation output and the voltage measurement input at the front of the device as shown in Figure 4. The plugs have a unique coding and only fit into their corresponding socket. Both cables come with bare wire ends to adapt them specifically to the connections of the cell to be measured. The polarity of the wires is specified with white for negative (-) and brown for positive (+).

Figure 4: Connections at the front

It must be ensured that a 4-point measurement is guaranteed so that no contact resistances of the connections are measured as illustrated in Figure 2. Figure 5 shows an example of a 4-point measurement on a pouch cell with Kelvin crocodile clamps. These allow for an easy setup of the experiments but are not recommended for long-term experiments.

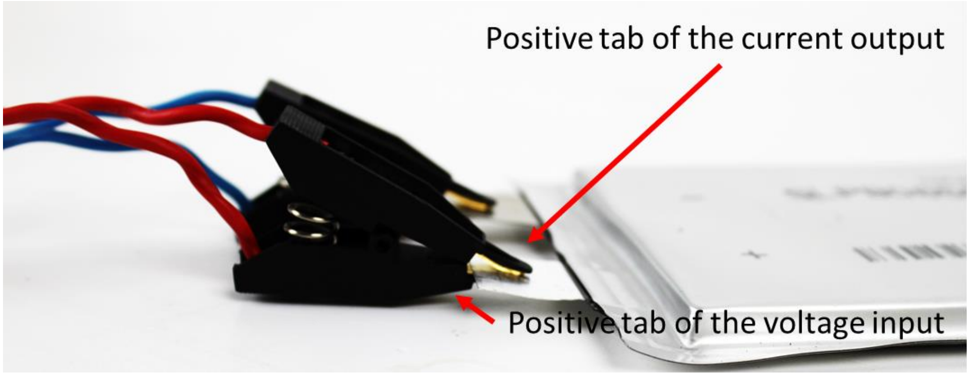
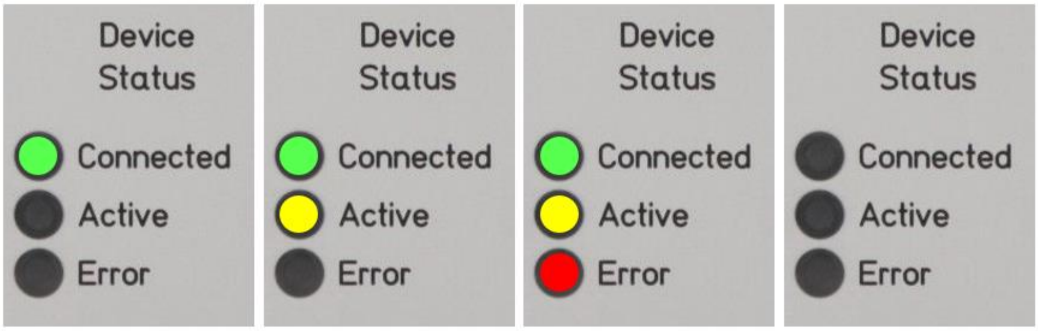


Figure 5: Kelvin crocodile clip connections

6. Device Power-Up

The device is powered on by toggling the power button on the front plate (2). Afterwards, the power Button will be illuminated with a green backlight and the three status LEDs (3) will sequentially turn on in order to indicate the successful initialization of the internal hardware.



7. Multiplexer Connection

In order to connect the multiplexer with the Inspectrum.C, the two devices need to be linked together with the four unique patch cables delivered with the multiplexer. Since the cables are coded, they only fit into their corresponding jacks. A simplified visualization of the multiplexer connection setup is given in Figure 6.

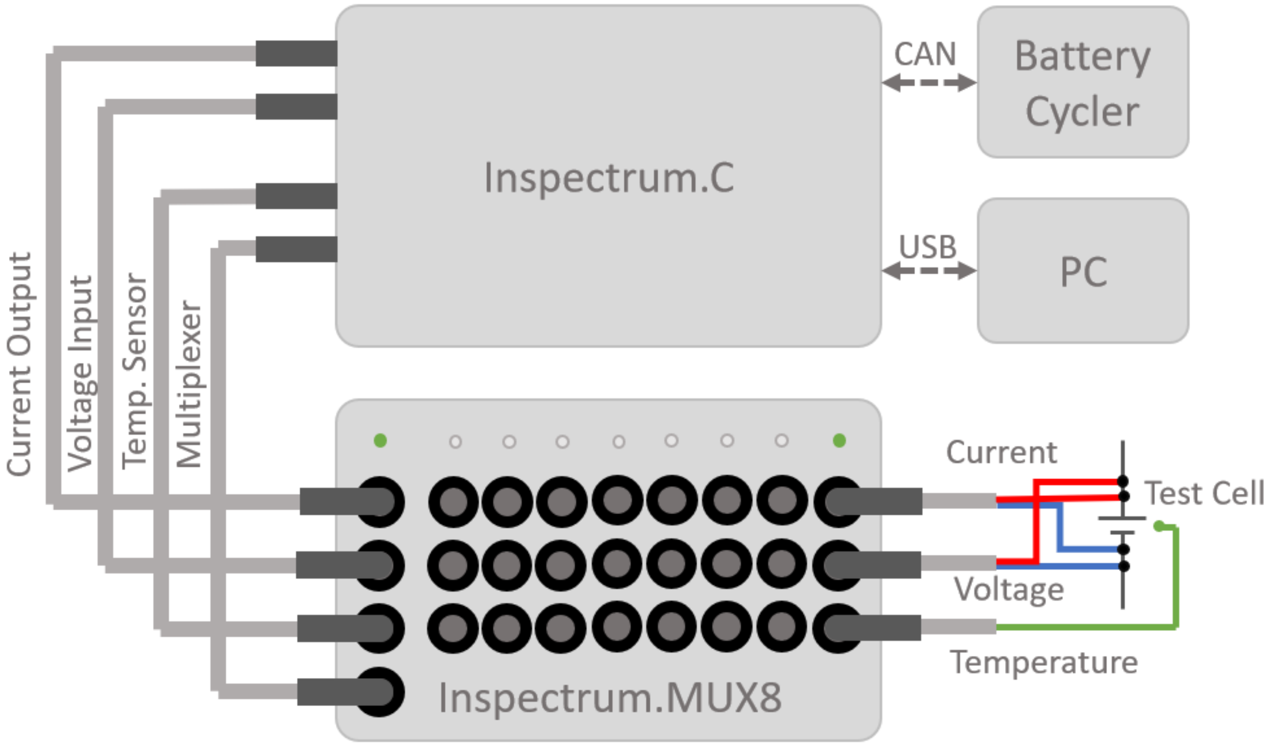


Figure 6: Diagram of the general connection setup of the multiplexer

It is important, that the multiplexer is connected before powering up the Inspectrum.C. Otherwise, the external device will not be recognized and a restart might be necessary. After powering up the Inspectrum.C, the “Status” LED of the multiplexer will light up red and switches over to green shortly after in order to indicate a successful connection. The battery connection is then executed in the same way as described in the chapter 5 and shown in Figure 6.

8. Software Installation and usage

For information regarding installation and usage of the Inspectrum.C control software (also called Inspectrum.Suite), please refer to the respective software user manual.

9. Maintenance

Calibration

The Device is factory calibrated on delivery. SAFION recommends recalibrating the device every 12 months. Please contact the SAFION customer service for more information about the calibration process.

Cleaning

Unplug the device from the power supply and disconnect any connected battery cell, before cleaning the device. It is recommended, to use a damp cloth for cleaning. Do not use any reactive cleaning agents or alcohols.